

Deepak Mallampati

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Kent, OH (open to London relocation) -- F-1 OPT, open to UK sponsorship

PROFESSIONAL SUMMARY

Applied AI engineer (4+ years) who ships production agentic and GenAI systems end-to-end -- comfortable owning customer outcomes from architecture through delivery. Built a **conductor-subagent agentic framework** in production healthcare (reduced token spend ~42% while sustaining task accuracy) and operated **25+ agents across 20+ teams at Vanguard scale** (100K+ requests/day, p95 ~800ms). Production track record: ~92% retrieval precision and 40% hallucination reduction on regulated-domain RAG over millions of documents. Bias to action, deep technical ownership, and comfort presenting architecture to VPs of Engineering -- the FDE profile.

CORE COMPETENCIES

Forward-Deployed Engineering . Customer-Embedded Delivery . Agentic LLM Systems . Production RAG . LLM Orchestration & Evals (RAGAS, BERTScore) . Python & TypeScript / FastAPI / React . Docker, Kubernetes, Multicloud Deploy (AWS, Azure, GCP) . End-to-End Ownership . Architectural Communication to VP-Level Stakeholders . Solo Ship Discipline

EXPERIENCE

Progress Solutions Inc.

Jul 2025 -- Present

AI Engineer -- USA | Healthcare Technology, AI Consulting

- Architected production **agentic LLM systems** on a conductor-subagent orchestration framework, decomposing complex objectives into context-scoped subagents -- reduced token consumption ~42% while sustaining accuracy across multi-step workflows.
- Engineered high-performance **RAG pipelines** over large-scale healthcare document corpora: dense vector retrieval + cross-encoder re-ranking lifted answer relevance and materially reduced hallucination on domain-specific queries.
- Established a structured **LLM evaluation and monitoring framework** (RAGAS, BERTScore, custom domain metrics) with latency/accuracy dashboards to benchmark iterations pre-deployment and surface regressions before release.
- Built fault-tolerant **automation infrastructure** (scheduled batch orchestration, exponential-backoff retry); reduced failed production runs ~60%.
- Optimized inference cost and latency via prompt compression, semantic caching, and model routing -- sustained SLA targets while reducing per-query overhead.
- Translated agentic capability for non-AI clinical stakeholders with scoped deliverables and eval-backed claims -- supported production adoption in HIPAA-conscious environments.

Vanguard (via consulting)

2023 -- 2025

AI / Platform Engineer -- USA Remote | Financial Services

- Operated **25+ agents across 20+ teams** in production at **100K+ requests/day** with **p95 ~800ms**; orchestration reliability + safeguards + observability sustained SLA across the surface.
- Built grounded GenAI services with FastAPI packaging, structured logging, and rollback gates -- cut post-deployment defects ~30%.
- Embedded with product/eng pods to translate AI capability into shipped features; owned architecture decisions and presented to senior stakeholders.

Kent State University -- ML & GenAI Research

Oct 2023 -- Jan 2024

Research Assistant -- USA | Privacy-Preserving ML, Long-Form GenAI

- Fine-tuned **Qwen3-27B via 4-bit QLoRA** on NVIDIA H100 with curated 361-example instruction dataset for schema-conditioned long-form generation (Hook -> Foreshadow -> Obstacle -> Amplifier -> Twist -> Payoff).
- Built privacy-preserving ML pipeline with k-anonymity (k=3), l-diversity (l=2), Laplace differential privacy -- cut re-identification risk from **3.38% to 0.32%** while holding **99%** of baseline RF AUC.
- Authored peer-review-ready IEEE-format paper documenting methodology and results.

Energy Solutions International (Emerson)

Jun 2022 -- Apr 2023

Database & DevOps Performance Engineer (Intern) -- Hyderabad, India | Enterprise ERP

- Built CI/CD pipelines (Jenkins) for the Synthesis Order-to-Cash ERP -- reduced deployment errors >**30%**, improved release cycle ~**35-40%**.
- Designed Grafana observability dashboards for production SQL systems -- cut incident recurrence ~**25%**, improved RCA speed ~**30%**.

PROJECTS

career-ops shipped -- in daily use

Self-built AI-powered job-search command center: portal scanner (zero-token Greenhouse/Ashby/Lever APIs), LLM-driven offer evaluator with structured A-H reports, automated PDF generator (Playwright + ATS-clean Times-Roman template), application tracker, and overnight CI/CD routine. **Demonstrates founder-energy end-to-end shipping** -- design, implementation, evaluation infra, and production CI all owned solo.

Node.js, Playwright, LLM tool-use, structured eval, CI/CD (GitHub Actions), Markdown-as-source

Agentic Healthcare Claims Processing & Fraud Risk Intelligence

Multi-agent pipeline (Intake -> Validation -> Consistency -> Duplicate Detection -> Fraud Risk) with **schema-validated JSON contracts between agents** to prevent cascading hallucinations. RAG-grounded CPT/ICD validation, ANN similarity for duplicate detection, explainable risk scoring with audit-ready reasoning traces.

Python, LangChain, vector search, FastAPI, Docker, regulated-domain governance

Story Director (QLoRA Fine-Tune of Qwen3-27B)

Schema-conditioned long-form generation via 4-bit QLoRA on a 361-example curated instruction dataset (NVIDIA H100). Demonstrated that **curated data beats compute** for targeted capability addition; produced a controllable narrative generator with consistent six-stage structure.

Hugging Face Transformers, PEFT/QLoRA, NVIDIA H100, instruction-tuning

EDUCATION

M.S. Computer Science -- Kent State University, Ohio, USA

Aug 2023 -- May 2025

TECHNICAL SKILLS

Languages & Frameworks: Python, TypeScript, JavaScript, Node.js, React, FastAPI, Flask, SQL (PostgreSQL, T-SQL, SQLite)

AI/LLM & Agentic: LLM APIs (Claude Sonnet, GPT-4o/5, Gemini, LLaMA, Ollama), LangChain/LangGraph, RAG, multi-agent orchestration, schema contracts, prompt engineering, structured outputs, LLM evals (RAGAS, BERTScore), guardrails, grounding

Vector Search: FAISS, Pinecone, Weaviate, hybrid + semantic search, cross-encoder re-ranking, semantic caching

Cloud / MLOps: AWS (Bedrock, SageMaker, Lambda, S3, EC2), Azure (OpenAI, AI Services), GCP Vertex AI, Docker, Kubernetes, MLflow, Weights & Biases, model monitoring

Forward-Deployed Tooling: Customer-embedded delivery patterns, eval-driven iteration, architectural communication to VP-level stakeholders, scoped pilots, end-to-end ownership

Regulated Domains: HIPAA-conscious governance, de-identification, audit trails, system-limitation documentation -- transferable to UK/EU compliance contexts

Work Authorization: US-based on F-1 OPT -- open to UK relocation with Skilled Worker visa sponsorship